Chapter 4: **Connectivity**

4.1 Cycles in 2-connected Graphs

We start with some useful facts about 2-connected graphs.

WHITNEY. *A graph with at least three vertices is 2-connected if and only if every pair of vertices are in a cycle.*

PROOF. (1) Assume every pair of vertices lie in a common cycle. Let v be a vertex and consider any two other vertices x and y. The condition implies there is a cycle containing x and y, and hence there are two paths between x and y with no internal vertex in common. The removal of a vertex v can affect only one of the paths. Thus v is not a cut-vertex. Since this is true for every vertex v, there does not exist a cut-vertex. That is, the graph is 2-connected.

(2) Assume the graph is 2-connected. Let x and y be distinct vertices. We claim that there is a cycle containing x. For, we observed earlier that being 2-connected implies every vertex has degree at least 2. If u_1 and u_2 are neighbors of x, then in $G - x$ there is still a path between u_1 and u_2; this combined with the u_1, x, u_2 path produces a cycle containing x.

Now, out of all cycles containing x, choose the cycle C such that the distance from y to the cycle is as small as possible. Suppose, for sake of contradiction, that y is not on C. Consider a shortest path from C to y; say it starts with vertices a and b (where possibly $b = y$). By the 2-connectendess assumption, there is a path from b to $C - a$ in the graph $G - a$. This path can be combined with part of the cycle C to produce a cycle C' that contains both x and b. But this cycle contradicts the original choice of C, since y is at least one closer to C'. Thus it must be the case that C contains y, as required. ⌑

Observe that if a graph is 2-connected then subdiving any edge yields a graph that is still 2-connected.

CYCLES THROUGH PAIRS. *Assume graph G has no isolates. There is cycle through each pair of vertices in G if and only if there is cycle through each pair of edges in G.*

Proof. (1) Assume G has a cycle through each pair of edges. Then to get cycle through specified pair of vertices just choose an edge incident with each vertex.

(2) Assume G has a cycle through each pair of vertices. By previous result this means G is 2-connected. For each of the specified pair of edges e and f, subdivide the edge with vertex v_e and v_f. The result is still 2-connected and so one can find a cycle containing v_e and v_f. This cycle necessarily uses the two edges incident with v_e and with v_f; so if we erase v_e and v_f we have a cycle using edges e and f. ⌑

4.2 Disjoint Paths

We say a collection of paths are ***internally disjoint*** if their internal vertices do not appear on any other path. For example, two vertices are joined by two internally disjoint paths if and only if they are in a common cycle.

Menger. *If x and y are nonadjacent vertices, then the minimum size of an x–y cut is equal to the maximum number of internally disjoint x–y paths.*

ProofSketch. If the graph consists of only x, y, and their neighborhoods, then use König-Egerváry to prove there is a matching between $N(x)$ and $N(y)$. Otherwise set things up to use induction. ⌑

Building on this, we have the following whose proof we omit:

Whitney. *A graph is k-connected if and only if there are k internally disjoint paths between every pair of vertices.*

A useful version (exercise):

Fan. *If a graph is k-connected, and $x, y_1, \ldots, y_k$ are disjoint vertices, then there are k internally disjoint paths P_i where P_i goes from x to y_i.*

From this we can prove:

Dirac. *For $k \geq 3$, if a graph is k-connected then every k vertices lie in a common cycle.*

Proof. Fix some subset B of k vertices. By the earlier result, we know there is a cycle containing at least two of B. Let C be a cycle that contains the maximum number of B. Suppose this is not all of B; that is, there is a vertex x of B not on C. Then, by the Fan theorem (or rather an extension thereof), there are $p = \min\{|C|, k\}$ internally disjoint paths from x to C, ending at different vertices of C. The p endpoints divide C into p segments. And by counting, there must be a segment that does not contain a vertex of B in its interior. The vertex x can be spliced into C replacing the segment, without losing any existing vertices of B, and thus we get a contradiction of the maximality of C. ⌑

There are similar results for edge-connectivity.

4.3 More

Graphs with bridges cannot have a strongly connected orientation. There is a converse whose proof we omit.

Robbins. *A graph has a strongly connected orientation if and only if it is 2-edge-connected.*

A network flow problem has a source, a sink and edge capacities, and the question is to maximize the flow from the source to the sink. Ford–Fulkerson: Max flow equals minimum cut. (Proof omitted. But sort of a generalization of Menger.) There is a connection with linear programming. Application: Network Flow can be used to compute the vertex and edge connectivity, and matching number.